

Maximum Mark: 15**MidSem-1****Time: 60 minutes**

1. (points) [3+1+1] Draw the fine structure of the Hydrogen atom for $n=3$ in comparison with the corresponding non-relativistic level (with proper labeling) and provide the reason for the order in splitting. Briefly discuss the degeneracy of the energy levels with and without fine structure effect.
2. (points) [2] Estimate the relative energy shift of $\Delta E/E_{1s}$ for the ground state of the Hydrogen atom where ΔE is the energy change due to the gravitational force and E_{1s} is the ground state energy of Hydrogen.
3. (points) [2] Argue and demonstrate explicitly that the fine structure corrections to the H-atom energy levels lead to a more tightly bound state (increase in binding energy) vis-a-vis the non-relativistic discription.
4. (3 points) Atoms are quantum systems. Prior to the discovery of the neutron, the neutrality of an atom necessitated that there should be $(A-Z)$ electrons in the nucleus of size $r \sim 10^{-14}$ m (Z : atomic number; A : atomic mass number). Show/demonstrate that such an atom is not possible.
5. (3 points) Write the generic Hamiltonian for an H atom. From it, explicitly derive the corresponding H in the centre of mass (centre of momentum) frame (show all steps explicitly). Write the total solution wavefunction for the latter (H in the CM frame).

Useful Results

- $R_{10} = 2 \left(\frac{1}{a_0} \right)^{3/2} e^{(-r/a_0)}$;
 $\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{(-r/a_0)}$;
 $\psi_{200} = \frac{1}{2\sqrt{2\pi}} \left(\frac{1}{a_0} \right)^{3/2} \left(1 - \frac{r}{2a_0} \right) e^{(-r/2a_0)}$;
- $\Delta E_{nj} = E_n^0 \frac{(Z\alpha)^2}{n^2} \left(\frac{n}{j+1/2} - \frac{3}{4} \right)$;
 $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$;
 $E_n^0 = -\frac{1}{2}\mu c^2 \frac{(Z\alpha)^2}{n^2}$
- mass of electron, $m_e \simeq 10^{-30}$ kg;
 speed of light, $c \simeq 3 \times 10^8$ m/s;
 mass of proton, $m_p \simeq 1.6 \times 10^{-27}$ kg;
 electric constant, $4\pi\epsilon_0 \simeq 10^{-10}$ F/m;
 electronic charge, $e \simeq 1.6 \times 10^{-19}$ C